Two Tiles for Every Stacking

The Endless Stack Cell: every close-packed stacking is a string of two layer tiles



DICTO (artist, Osaka, Japan) · contact@dictospheres.com · https://dictospheres.com · ORCID 0009-0001-2620-2036
Paper #75 · Version 1 · 29 October 2026 · DICTO DOI: 10.5281/zenodo.23737639 ·
Licence CC BY-SA 4.0 · Scripts, exact checks and data: https://github.com/DICTOR-Master/kaleidohedra/tree/fcb20088/studies/2026-10-29-endless-stack-cell (research branch, commit fcb20088); a snapshot is attached to this record
The idea is DICTO's (29 October 2026): that the pieces of the Rule of Halves fill one unit and join across units without end, as the Euclid–Kepler–Pacioli cell network does. Computation, exact proof and drafting with Claude (Anthropic).

Abstract. In the Rule of Halves (#57) every hole of every close-packed stacking gets a piece, and each piece is made from the pieces of the two pure stackings, cubic and hexagonal. Here we cut space into layer tiles: the tile of a layer is the share of space of that layer, made of four pieces, the top half of the octahedral piece below it, the two tetrahedral pieces next to it and the bottom half of the octahedral piece above it. We prove, exactly and along the whole weight walk of #57, that there are only two tiles. Every layer tile is carried onto the c-tile of the cubic stacking or the h-tile of the hexagonal stacking by one motion, all four pieces at once, and which of the two is decided by the layer's own type. Any tile sits on any tile, since the joint between two layers has the same shape for both types. So every close-packed stacking, periodic or not, is a string of two tiles, and a string of two tiles in any order is a stacking. The tiles meet on flat planes for weights up to $\lambda = -7/24$ and interlock above it, their tetrahedral pieces reaching into the neighbouring layer. At each weight all stackings together use five whole shapes.

Stack layers of balls and the gaps between them can be shared out among the holes. Paper #57 found that every piece is built from two kinds of half. Here the halves are packed into layers, and only two kinds of layer come out, one for each way a layer can sit between its neighbours. Pick any sequence of the two and you have built a crystal stacking; the pieces fit at every joint, endlessly.

1. Where it started

The Euclid–Kepler–Pacioli cell (#8 [9]) holds all five Platonic solids in one unit, and copies of the unit join across their faces so that the cycle of solids runs on from cell to cell without end. DICTO proposed that the pieces of the Rule of Halves (#57) behave in the same way: that they fill one unit and join across units, cycling without end. The natural unit is one layer of balls with its share of the holes on either side. This paper proves that the idea holds exactly, and finds the shape of the two units.

2. The set-up

As in #57: unit balls sit in close-packed layers (triangular lattices of spacing 1) at heights kd, $d = \sqrt{(2/3)}$; layer k sits on one of the three positions A, B, C, with no two consecutive layers on the same position. Layer k is *h* if layers $k - 1$ and $k + 1$ sit on the same position and *c*

otherwise. Between layers k and k + 1 lie an octahedral hole at height (k + ½)d and two tetrahedral holes, at (k + ¼)d and (k + ¾)d. One weight λ, given to the tetrahedral holes (the octahedral holes have weight 0), decides the power diagram of the holes, whose cells are the pieces; the walk is $-11/24 \le \lambda \le 3/8$. By #57, a tetrahedral piece is the cubic or hexagonal tetrahedral piece named by its nearest layer, and an octahedral piece cut by the horizontal plane through its hole has its bottom half named by the layer below and its top half by the layer above.

Definition. The *tile* of layer L, for one ball of the layer, is made of four pieces: the top half of the octahedral piece of the gap below (hole at (L − ½)d), the tetrahedral pieces at (L − ¼)d and (L + ¼)d, and the bottom half of the octahedral piece of the gap above (hole at (L + ½)d). The *c-tile* and *h-tile* are the tiles of the pure cubic stacking ABC… and the pure hexagonal stacking AB…. Every piece and half of the diagram belongs to exactly one tile, so the tiles of all balls of all layers fill space without overlap.

3. Main theorem

> **Theorem (two tiles).** For every close-packed stacking and every λ with $-11/24 \le \lambda \le 3/8$, the tile of each layer L is carried onto the c-tile, if L is c, or onto the h-tile, if L is h, by one motion that maps the layer to itself: a symmetry of the triangular lattice about the vertical axis through the ball, followed by a translation. The same motion carries all four pieces.

Corollaries. (i) *Any tile sits on any tile.* The face where two tiles meet is the section of an octahedral piece by the horizontal plane through its hole; for every λ this section has one shape, shared by the c- and h-tiles. Every sequence of c- and h-tiles is therefore a stacking, and every stacking is such a sequence. (ii) *Flat or interlocking.* For $\lambda \le -7/24$ every tile lies between the planes through the octahedral holes above and below, and the tiles meet on flat planes; for $\lambda > -7/24$ the tetrahedral pieces cross these planes and the tiles interlock. The value −7/24 is the same for both tiles and is not a stop of the walk. (iii) *Five shapes.* At each λ all close-packed stackings together use five whole pieces: the c- and h-tetrahedral pieces and three octahedral pieces, with two c-halves, two h-halves, or one of each.

The Rule of Halves matched each piece separately, each with its own motion. The theorem here is stronger: one motion places the whole tile, so the tile is a rigid unit that can be picked up and stacked.

4. Proof

Step 1: seven layers decide. The two octahedral halves and two tetrahedral pieces of tile L belong to the gaps L − 1 and L. By Step 1 of #57 the pieces of a gap depend only on the six layers around it, so the tile depends only on the layers L − 3, …, L + 3. Up to relabelling A, B, C these are fixed by the types of the five inner layers L − 2, …, L + 2: 32 contexts. For each we took a periodic stacking containing the seven layers; the 32 stackings are listed in prove.out [S].

Step 2: the stretches. For each context we found exactly, with the census engine of #53 and #57, every λ at which one of the four cells of the tile, or of the pure tile of the same type, changes its faces. Between two such breakpoints every corner of every cell is an affine function of λ, and the corners of a half, where an edge crosses the cutting plane, are rational functions of λ. On each sub-interval we found the motion at one sample λ, applied it to the

tile, and compared the result with the pure tile as rational functions of λ (computer algebra, exact): they are identical, so the tiles are equal on the whole sub-interval. All 32 contexts, every sub-interval: equal.

Step 3: the breakpoints. At each breakpoint the tile was computed directly, in rational arithmetic, and the same comparison holds. □

Corollary (i). In a stacking with a c-layer below an h-layer, the octahedral piece between them has a c-half below and an h-half above. Its section by the plane through its hole is the top face of the c-half and the bottom face of the h-half at once. By the theorem every c-tile's top face is a copy of this c-face and every h-tile's bottom face a copy of this h-face, and they are the same shape. The same holds the other way up, and for c on c and h on h trivially. **Corollary (ii).** On each sub-interval the heights of the tetrahedral corners are affine in λ; solving where they meet ±½ (in layer units) gives $\lambda = -7/24$ in all 32 contexts, and no other crossing. **Corollary (iii).** In a c-layer the ball is a centre of symmetry of the cubic stacking, which swaps the two tetrahedral pieces of the tile and its two octahedral halves; in an h-layer the layer plane is a mirror of the hexagonal stacking, which does the same. So each tile holds one tetrahedral shape and one half shape, and the octahedral pieces are made of two halves of these four kinds: c + c, h + h or c + h (the mixed pieces of #57, Chucky and Fang among them). □

> **Computational methods.** Python with exact rationals (fractions) and sympy, using the census engine of #53 [2] unchanged; floating point only proposes candidate corners, which are then solved and checked exactly. Sample check (tiles.py): 32 contexts × 10 values of λ, 320 of 320 tiles matched by one motion. Proof (prove.py): 32 of 32 contexts, every sub-interval and breakpoint. Run on the dicto-big-brother node, 29 October 2026.

5. The two tiles

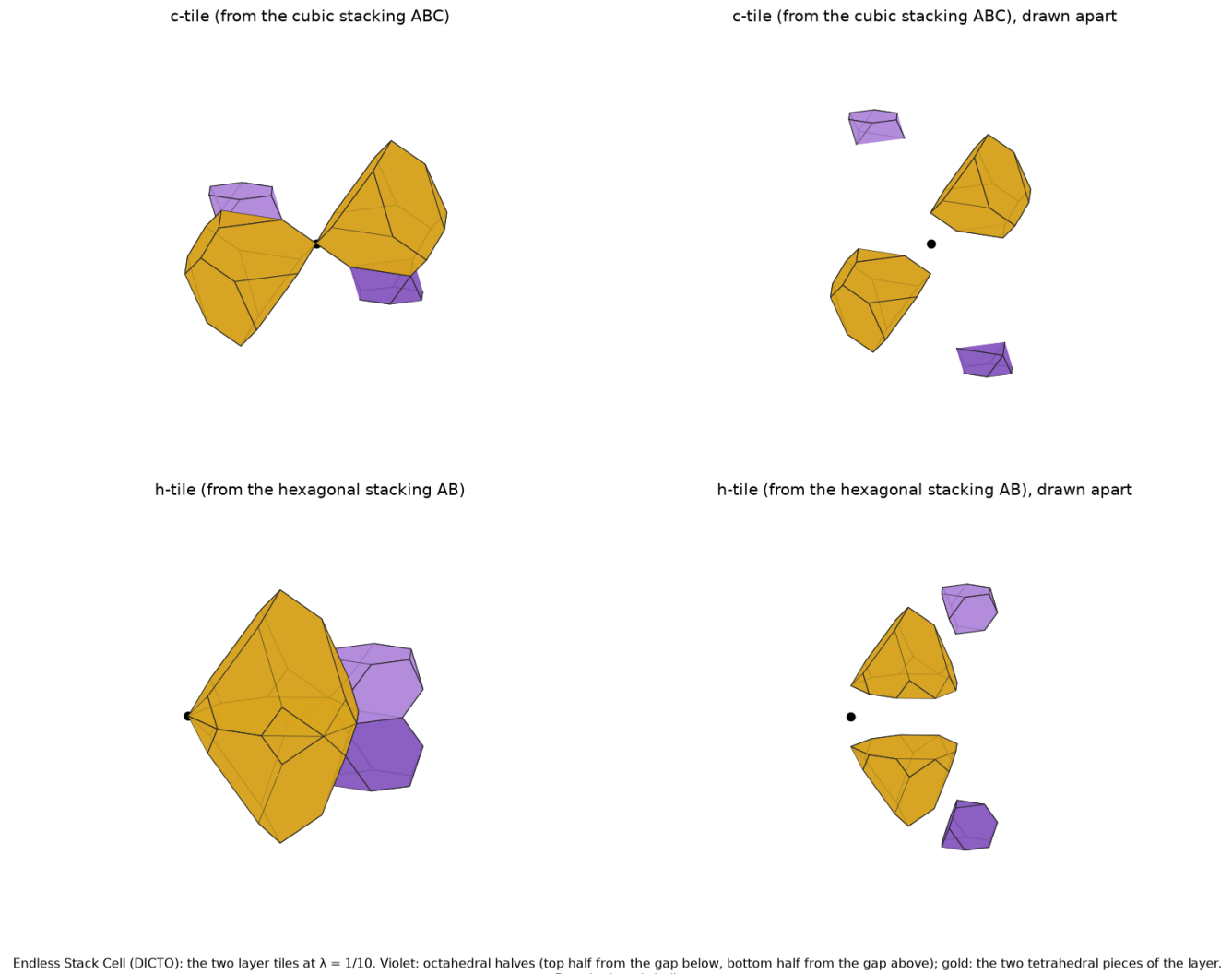


Figure 1. The c-tile (top) and h-tile (bottom) at λ = 1/10, whole (left) and drawn apart (right). Gold: the two tetrahedral pieces, meeting at the layer's ball (black dot); light violet: the bottom half of the octahedral piece above; dark violet: the top half of the octahedral piece below. In the h-tile the two halves lie on one vertical line, because the layers above and below sit on the same position; in the c-tile they are set apart, because those layers sit on different positions. That is the whole difference between the two tiles.

A tile is one ball's share of its layer; side by side, copies of a tile fill the layer as a slab whose upper and lower faces are flat for $\lambda \le -7/24$ and toothed above. At $\lambda = 1/10$ the tetrahedral pieces reach a full layer up and down, to the height of the next layer's balls, so each layer keys into the next (Figure 2).

So there is a cubic tile and a hexagonal tile, and either fits on the top or bottom face of either. Sideways a layer is made of one tile only, since all its balls have the same type. The type of a tile does not limit what goes above it; it decides where the next layer sits: above a c-tile the next layer takes the third position, above an h-tile it returns to the position of the layer below.

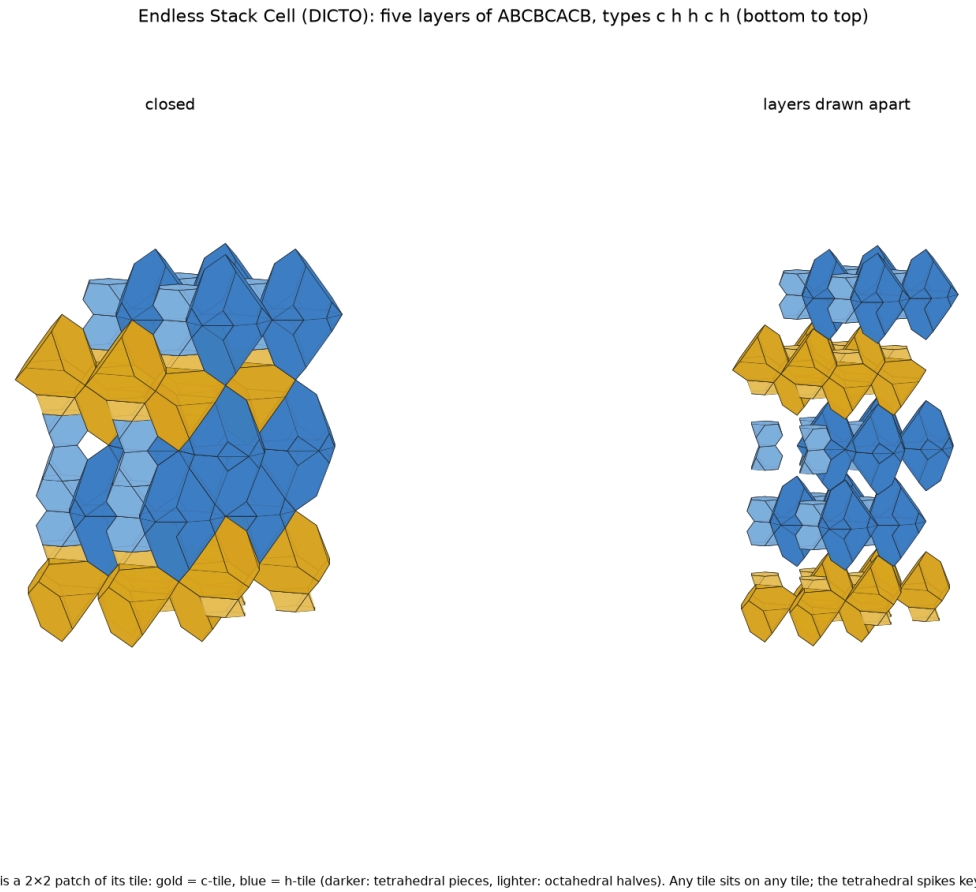


Figure 2. Five layers of the stacking ABCBCACB, types c h h c h from the bottom, at λ = 1/10; each layer a 2 × 2 patch of its tile, gold for c-tiles and blue for h-tiles (darker: tetrahedral pieces; lighter: octahedral halves). Left closed, right with the layers drawn apart.

6. What is known, and what was not found

Known and credited. Close-packed stackings, polytypes and the h/c notation [4]; power diagrams [5]; the pieces and the Rule of Halves [1] on the walk of #53 [2], with the twinned octet pieces of #56 [3]. Order–disorder (OD) theory, from Dornberger-Schiff, describes polytypic crystals as stacks of layers of one or more kinds, any two neighbouring layer pairs being geometrically equivalent (the vicinity condition) [6]: the general frame in which a stacking is a string of layers. Tilings of space by regular tetrahedra and octahedra, including non-periodic ones obtained by sliding along planes, are enumerated by Gabbrielli, Jiao and Torquato [7]; the unit cell of the octet as a rhombohedron of two tetrahedra and one octahedron, and its layered sections, are in Lalvani [8]. The Euclid–Kepler–Pacioli cell network is DICTO's [9].

Stated here. The layer tiles built from the power-diagram pieces of the holes; the proof that there are only two, placed each by one motion, for all stackings and the whole walk; the shared joint; the switch from flat to interlocking at λ = −7/24; the five shapes. We found no earlier statement of tiles of this kind.

> **Sources checked** (29 October 2026). Reference library, by meaning search (four searches: unit cell of tetrahedra and octahedra halves matching across cell faces; half-octahedron pieces filling close-packed stackings; polytypes as tilings with layer-to-layer matching rules; periodic unit-cell tilings continuing across cells). Read: R. Gabbrielli, Y. Jiao and S. Torquato 2012 (pp. 1–4); H. Lalvani 1990, NASA TM-102635 (pp. 5–6); Wikipedia, Tetrahedral-octahedral honeycomb. Seen in the hits and judged not relevant

(packings and non-periodic monotiles): Y. Kallus, V. Elser and S. Gravel 2009; S. Torquato and Y. Jiao 2009; P. Kramer 1982; D. Smith, J. S. Myers, C. S. Kaplan and C. Goodman-Strauss 2023; I. Tsiokos 2026. Web searches: OD theory and close-packed polytypes (IUCr Acta B 2025, dk5140, read in summary); slab tiles of tetrahedra and octahedra in close packing (arXiv 1407.0692, which splits fcc and hcp between layers into tetrahedra and octahedra, one per triangle); power and radical Voronoi diagrams of interstitial sites (arXiv 2010.01354; Scientific Data 2020, CAVD), which have no layer tiles. Our own papers #57 and the book The Laws of Stacking: no layer tiles.

7. Where the results stop

(1) The theorem covers the walk of #57, $-11/24 \le \lambda \le 3/8$, with one weight for all tetrahedral holes; at the ends some pieces shrink to points and the tiles are read as limits. (2) With different weights for the two kinds of tetrahedral hole the tiles may split into more kinds; not tested. (3) The tiles are geometric; which of them a real crystal prefers is the subject of the Hexagonality Law papers, not of this one.

8. Conclusion

Every close-packed stacking, at every weight, is a string of two tiles, and any string of the two tiles is a stacking. Each tile is one rigid unit of four pieces, decided by its own layer alone; the tiles meet on flat planes up to $\lambda = -7/24$ and interlock beyond it. As DICTO proposed, the pieces fill one unit and join across units without end.

References

[1] DICTO, "The Rule of Halves: the Holes of Every Close-Packed Stacking", paper #57, DICTO DOI: 10.5281/zenodo.23647898 (2026).
[2] DICTO, "The Global Web of Bonds: How Bonds Hand Over Across Crystal Networks", paper #53, DICTO DOI: 10.5281/zenodo.23631140 (2026).
[3] DICTO, "Sonsam and Lahdeli: the Delian Bond on the Twinned Octet", paper #56, DICTO DOI: 10.5281/zenodo.23635823 (2026).
[4] International Tables for Crystallography, Vol. C, ch. 9.2 (2006), section 9.2.1, close-packed structures and the h/c notation.
[5] F. Aurenhammer, "Power diagrams: properties, algorithms and applications", SIAM Journal on Computing 16 (1987) 78–96, doi:10.1137/0216006.
[6] K. Dornberger-Schiff and H. Grell-Niemann, "On the theory of order–disorder (OD) structures", Acta Crystallographica 14 (1961) 167–177; and the account of OD theory in Acta Crystallographica B 81 (2025), https://journals.iucr.org/b/issues/2025/06/00/dk5140/.
[7] R. Gabbrielli, Y. Jiao and S. Torquato, "Families of tessellations of space by elementary polyhedra via retessellations of face-centered-cubic and related tilings", Physical Review E 86 (2012) 041141, doi:10.1103/PhysRevE.86.041141.
[8] H. Lalvani, "Transformational part-count in layered octahedral-tetrahedral truss configurations", NASA Technical Memorandum 102635 (1990).
[9] DICTO, "The Euclid–Kepler–Pacioli Cell", paper #8, DICTO DOI: 10.5281/zenodo.23284106 (2026).
[S] DICTO, studies/2026-10-29-endless-stack-cell (commit fcb20088): tiles.py, prove.py, render_tiles.py, render_stack.py; outputs tiles.out, prove.out; README.md (prior art and results).